

ENGINEERING SPECIFICATION • SOURCE OF TRUTH

The Business Intelligence Engine.

How Auxion actually thinks. Every AI capability inside the platform — from crawling to reasoning to proposal generation to continuous monitoring — originates from this specification. It follows PDFs 00–26 and contradicts none of them: PDF 26 governs scan behavior, the earlier documents govern design language.

18 SECTIONS

8 ENGINE LAYERS

PROVIDER-AGNOSTIC

EVIDENCE-DRIVEN

01 VISION

6 VERBS

Auxion is not a website scanner. It is an AI business intelligence platform. A URL is only the entry point; the engine's job is to understand a business, reason about it against its market, and drive it forward on a loop. Six verbs define the mandate.

01

Observe

Crawl the public surface and every connected source. Capture raw evidence without interpretation.

02

Understand

Normalize raw evidence into a structured business profile the system can reason over.

03

Reason

Weigh evidence against benchmarks and competitors to establish what is true, and why.

04

Recommend

Convert findings into scored, evidence-linked moves with impact, difficulty, and cost.

05

Transform

Package approved moves into a proposal, timeline, and implementation plan.

06

Continuously Improve

Re-scan on schedule, detect change, and keep the Business Health Index current.

02 CORE PHILOSOPHY

7 LAWS • NON-NEGOTIABLE

These laws bind every layer. No feature may violate them. They are what makes Auxion trustworthy in front of a client.

Evidence before reasoning

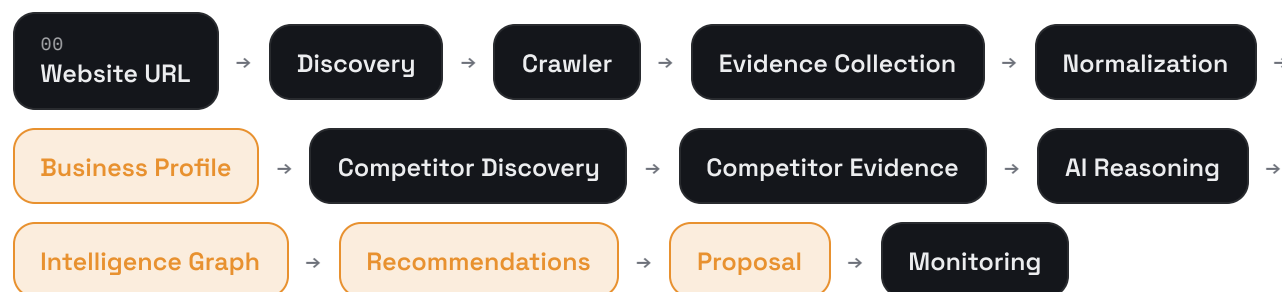
No conclusion is formed before its supporting evidence is collected, classified, and timestamped.

Never hallucinate	The engine reports only what evidence supports. Absence is stated plainly, never invented to fill a gap.
Cite everything	Every recommendation references the exact evidence trace that produced it — inspectable end to end.
Scores explain themselves	Every score exposes its inputs, weighting, and coverage. Nothing is a bare number.
Confidence is mandatory	Every conclusion carries a confidence value derived from the strength of its evidence.
Impact is explicit	Every recommendation states projected impact, difficulty, cost, and expected ROI up front.
Human in control	No black box. The operator can inspect, override, and approve every step before it reaches a client.

03 BUSINESS INTELLIGENCE PIPELINE

13 STAGES

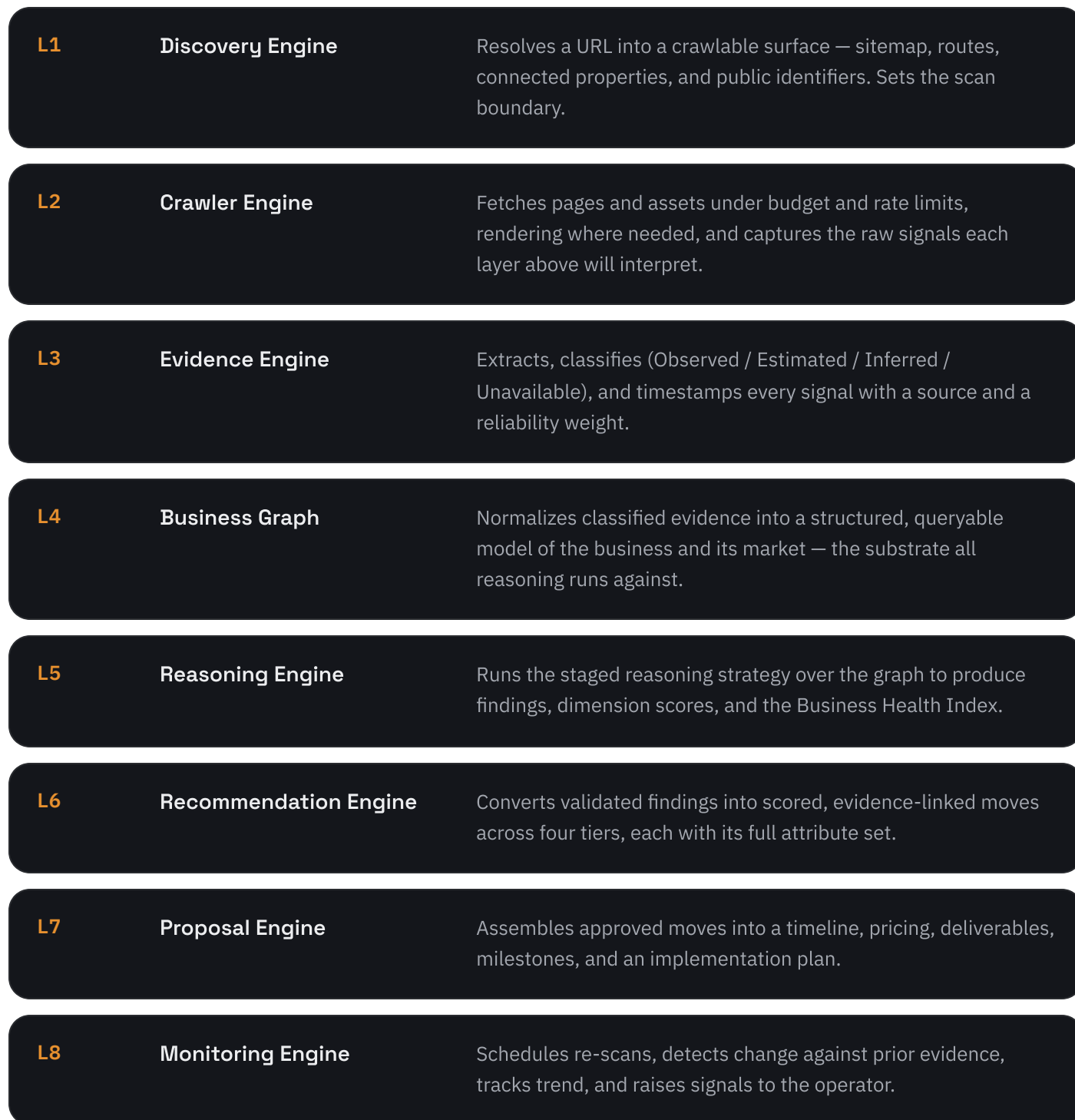
A single directed flow from raw URL to a monitored, improving account. Amber nodes are produced knowledge artifacts; neutral nodes are processes that transform them.



04 ENGINE LAYERS

8 LAYERS · COMPOSABLE

Eight independently testable layers. Each has one responsibility, a defined input and output, and can be replaced without disturbing its neighbors.



05 EVIDENCE SOURCES

19 SOURCES · 4 STATES

Every signal enters the engine tagged with one of four states. The state travels with the signal through every downstream layer and appears verbatim in the client-facing report — the system never presents an inference as an observation.

● Observed · directly measured ● Estimated · modeled from data ● Inferred · reasoned, not seen

● Unavailable
 · not yet accessible

Source	Default State	Note
Website	● Observed	Root domain, structure, and technology surface.
Pages	● Observed	Content, metadata, and internal linking per page.
SEO	● Observed	Titles, schema, indexability, and keyword surface.
Performance	● Observed	Load timing, weight, and rendering path.
Accessibility	● Observed	Contrast, semantics, and assistive-tech support.
Security	● Observed	TLS, headers, and exposed misconfiguration.
Brand	● Inferred	Positioning and tone reasoned from public surface.
Forms	● Observed	Capture points, fields, and conversion paths.
Analytics	● Unavailable	Requires a granted property; Unavailable until connected.
Social Media	● Observed	Public profiles, activity, and audience signal.
Google Business	● Observed	Listing completeness, categories, and hours.
Reviews	● Observed	Volume, rating, and sentiment across platforms.
Competitors	● Estimated	Discovered set scored on the same dimensions.
Industry Benchmarks	● Estimated	Category baselines the client is measured against.
Public APIs	● Observed	Third-party data where a public endpoint exists.
Manual Input	● Observed	Operator-supplied facts, treated as direct evidence.
Client Documents	● Unavailable	Unavailable until the client provides them.
Existing CRM	● Unavailable	Unavailable until a connection is authorized.
Historical Scans	● Observed	Prior evidence for this account, used for trend.

06 AI ORCHESTRATION

NO VENDOR LOCK-IN

The engine treats AI providers as interchangeable resources behind one internal interface. No vendor is hardcoded anywhere in the architecture. The providers below are examples only — added, removed, or reordered without changing a single downstream layer.

• OpenAI • Claude • Gemini • DeepSeek • Future providers

EXAMPLES ONLY · NEVER HARDCODED · SELECTION IS RUNTIME

A router selects a provider per task on these criteria:

Task

What the stage needs — extraction, reasoning, or writing.

Latency

How fast a result is required at this stage.

Cost

Price per token against the scan budget.

Reasoning quality

Depth required for the specific decision.

Context size

How much evidence must fit in one window.

Availability

Live health and rate-limit headroom.

Fallback

The ordered alternates if the first choice fails.

07 REASONING STRATEGY

6 STAGES

Reasoning runs as an ordered pipeline, not a single prompt. Each stage has a bounded responsibility and hands a validated artifact to the next.

01

Planner

Reads the profile and coverage, then sequences which questions to answer and which evidence each one requires.

02

Research

Fills the evidence gaps the planner flagged — requesting sources, or classifying them Unavailable when they cannot be reached.

03

Evidence Validation

Cross-checks signals for agreement, freshness, and reliability before any of them are allowed to inform a score.

04

Recommendation

Derives scored moves from validated findings, each carrying its full attribute set and evidence link.

05 Executive Summary

Renders findings into plain-language priorities for the client without losing the underlying evidence trace.

06 Proposal Writing

Turns approved moves into a client-ready proposal in the Auxion voice — state first, specific, never hyped.

08 CONFIDENCE MODEL

6 FACTORS

Every score and conclusion carries a confidence value. It is computed, not asserted, from six factors.

Confidence	The composite 0–100 value attached to every score and conclusion the engine emits.
Evidence Coverage	The share of the evidence a conclusion requires that was actually collected.
Source Reliability	Each source weighted by trustworthiness and how directly it observes the fact.
Freshness	A decay applied as evidence ages past its useful window for the dimension.
Agreement	The degree to which independent sources corroborate the same signal.
Completeness	Whether the profile has enough breadth to support the conclusion at all.

SCORING

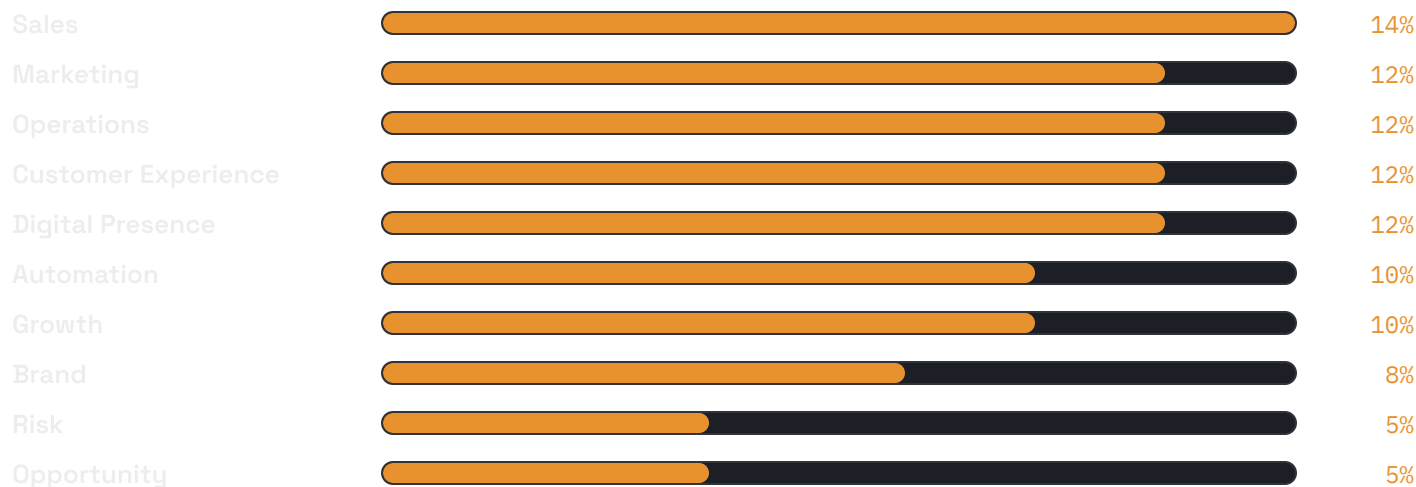
Confidence = f(coverage × reliability × freshness × agreement × completeness)

Any factor near zero caps the composite. When evidence is thin, the engine reports low confidence and states what is missing — it never raises confidence to fill a gap.

09 BUSINESS HEALTH INDEX

10 DIMENSIONS • 100%

The Index is a single 0–100 figure composed of ten weighted dimensions. Weights sum to 100 and are fixed per this specification; each dimension is itself evidence-backed and carries its own confidence. Risk is scored inversely — exposure lowers it.



10 COMPETITOR INTELLIGENCE

8 SIGNALS

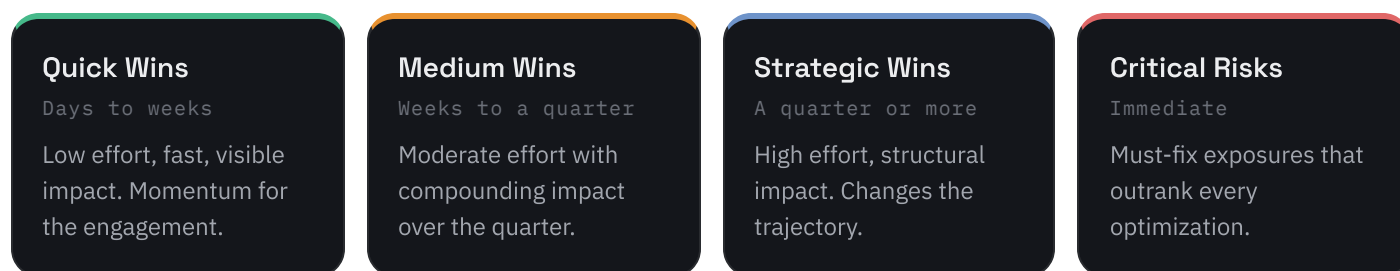
Competitors are scanned on the same evidence dimensions as the client, so every comparison is like-for-like and carries the same four-state honesty.

Discovery	Identify the competitive set from market, category, and observed positioning.	Ranking	Order competitors by relevance and overlap with the client's actual market.
Comparison	Evaluate each competitor on the same evidence dimensions as the client.	Gap Analysis	Surface where the client trails or leads on each dimension, with evidence.
Opportunity Matrix	Plot gaps by impact against effort to expose the highest-value moves.	Market Position	Locate the client within the competitive set on the dimensions that matter.
Risk	Flag competitor advantages that actively threaten the client's position.	Differentiators	Isolate where the client is defensibly ahead and worth protecting.

11 RECOMMENDATION FRAMEWORK

4 TIERS

Findings become moves, sorted into four tiers. Critical risks always outrank optimization regardless of effort.



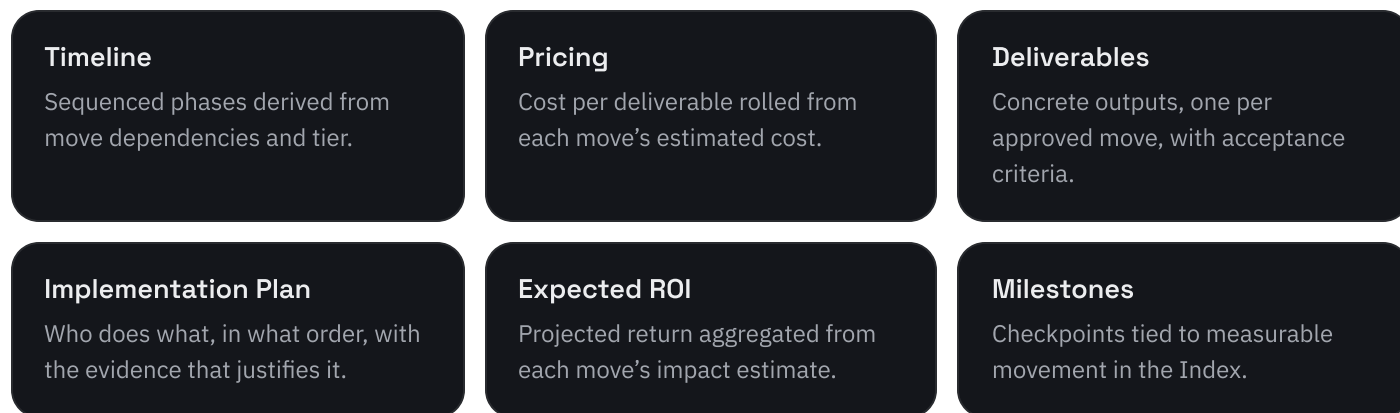
EVERY RECOMMENDATION CARRIES SEVEN ATTRIBUTES



12 PROPOSAL GENERATOR

6 PARTS

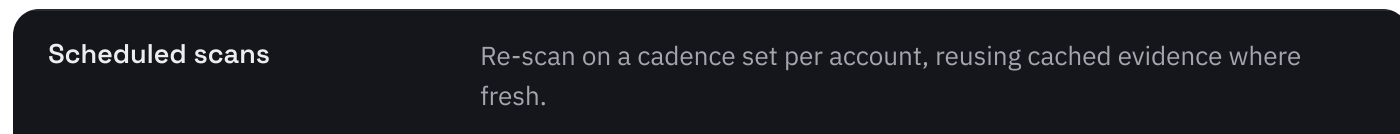
Approved moves assemble into a client-ready proposal. Nothing is written that isn't backed by an approved, evidence-linked move — the proposal inherits the whole evidence trace.



13 CONTINUOUS MONITORING

6 CHANNELS

A scan is not a one-time event. Once an account is live, the engine re-scans on schedule, watches for change, and keeps the Index current.



Change detection	Diff new evidence against the prior graph to find what actually moved.
Trend analysis	Track each dimension and the Index over time, not just the latest value.
Alerts	Raise a signal to the operator when a dimension crosses a threshold.
Weekly summaries	A concise digest of change, priorities, and open moves.
Monthly reports	A full re-render of the report with trajectory against target.

14 BACKGROUND PROCESSING

8 CONCERNS

A scan is long-running work. It runs asynchronously against a queue so the surface stays responsive and any stage can fail and recover in isolation.

Queue	Every scan and stage is a durable job, ordered and inspectable.	Workers	Stateless workers pull jobs and scale horizontally with load.
Retry	Transient failures retry with backoff before a job is marked failed.	Timeout	Every stage has a hard ceiling; a slow source degrades, never blocks.
Caching	Fresh evidence is reused across stages and scans to avoid re-fetching.	Rate limiting	Per-source and per-provider limits keep the engine a good citizen.
Provider fallback	A failed AI provider fails over to the next in the ordered set.	Parallel execution	Independent stages run concurrently; the graph enforces order only where needed.

15 COST MANAGEMENT

7 LEVERS

Intelligence has a marginal cost. Every scan operates inside a budget, and the engine spends tokens and API calls deliberately.

LLM budgeting	Each scan holds a token budget; the router spends against it per stage.	API budgeting	Third-party calls are metered and capped per account and per scan.
Caching	Reuse fresh evidence instead of re-fetching or re-reasoning it.	Reuse	Shared benchmarks and competitor data amortize across accounts.
Token optimization	Send only the evidence a stage needs — trimmed, not the whole graph.	Incremental scanning	Re-scan only what changed since the last run, guided by change detection.
Partial rescans	Refresh a single dimension or source without re-running the whole pipeline.		

16 SECURITY

7 CONTROLS

Evidence is a client's business laid bare. It is isolated, owned, encrypted, and auditable at every hop — including across AI providers.

Tenant isolation	Every account's evidence is isolated; no scan can read another's data.
Evidence ownership	Evidence belongs to the client; it is deletable and portable on request.
Encryption	Evidence is encrypted in transit and at rest across every layer.
Secrets	Provider keys and credentials live in a vault, never in code or logs.
Prompt protection	System prompts and reasoning strategy are not exposed to clients or providers.
Provider isolation	No provider retains client evidence; requests are scoped and stateless.
Audit logs	Every access, decision, and override is logged and attributable to an operator.

17 FUTURE EXPANSION

8 MODULES

The same engine — discovery, evidence, reasoning, recommendation — generalizes beyond the website. Each future module is a new evidence domain plugged into the existing layers, not a new engine.

Email Intelligence

Reason over email as an evidence domain — deliverability, engagement, cadence.

Sales Intelligence

Pipeline, conversion, and rep performance scored on the same model.

CRM Intelligence

Data health, coverage, and process gaps from the connected CRM.

Financial Intelligence

Margin, cash, and unit economics reasoned from financial evidence.

HR Intelligence

Hiring, retention, and org structure as measured dimensions.

Supply Chain Intelligence

Supplier, inventory, and logistics evidence and risk.

Document Intelligence

Extract structured evidence from contracts, decks, and reports.

Meeting Intelligence

Turn recorded meetings into evidence, decisions, and moves.

18 ENGINEERING PRINCIPLES

8 PRINCIPLES

The invariants every contributor holds. If a design decision violates one of these, it is wrong regardless of how well it performs.

01

No vendor lock-in — any provider, source, or store can be replaced.

02

Everything provider-agnostic — AI sits behind one internal interface.

03

Everything evidence-driven — no output without a cited source.

04

Every capability composable — layers combine without special cases.

05

Every stage independently testable — bounded input, bounded output.

06

Every decision explainable — the reasoning trace is always available.

07

Everything observable — every job, call, and stage emits telemetry.

08

Everything measurable — quality, cost, and confidence are all tracked.